Warm up about tensors and all that (Ch 1.4 of Brau) 10/15

Antisymmetric tensor ε_{iklm} : components ± 1 when all 4 indices are different, 0 when at least two coincide.

$\varepsilon_{0123} = -1$ and all other components are obtained by permutations — odd permutations change sign $(+1)$
even do not change (-1).

and $\varepsilon^{0123} = +1$.

Invariant $\varepsilon_{iklm} F^{ik} F^{lm} = \varepsilon^{iklm} F_{ik} F_{lm}$

$$F^{ik} = \frac{\partial A^k}{\partial x_i} - \frac{\partial A^i}{\partial x_k}$$

$$\varepsilon_{iklm} \left(\frac{\partial A^k}{\partial x_i} - \frac{\partial A^i}{\partial x_k} \right) \left(\frac{\partial A^m}{\partial x_l} - \frac{\partial A^l}{\partial x_m} \right) =$$

$$= \varepsilon_{iklm} \frac{\partial A^k}{\partial x_i} F^{lm} - \underbrace{\varepsilon_{iklm} \frac{\partial A^i}{\partial x_k}}_{} F^{lm} =$$

$$- \varepsilon_{kilm} \frac{\partial A^i}{\partial x_k} = - \varepsilon_{i'k'lm} \frac{\partial A^{k'}}{\partial x_{i'}} =$$

change $i \leftrightarrow k$

rename $k = i'$, $i = k'$

$$= - \varepsilon_{iklm} \frac{\partial A^k}{\partial x_i}$$

drop primes.

$$= 2 \varepsilon_{iklm} \frac{\partial A^k}{\partial x_i} F^{lm} = 2 \frac{\partial A^k}{\partial x_i} \underbrace{\varepsilon_{iklm} \left(\frac{\partial A^m}{\partial x_l} - \frac{\partial A^l}{\partial x_m} \right)}_{} =$$

by same logic, $2 \varepsilon_{iklm} \frac{\partial A^m}{\partial x_l}$

$$= 4 \varepsilon_{iklm} \frac{\partial A^k}{\partial x_i} \frac{\partial A^m}{\partial x_l}$$

$$\Rightarrow \varepsilon_{iklm} F^{ik} F^{lm} = 4\, \varepsilon_{iklm} \frac{\partial A^k}{\partial x_i} \frac{\partial A^m}{\partial x_l} =$$

$$= 4 \frac{\partial}{\partial x_i}\left(\varepsilon_{iklm} A^k \frac{\partial A^m}{\partial x_l}\right) - 4\, \varepsilon_{iklm} A^k \frac{\partial^2 A^m}{\partial x_i \partial x_l}$$

But $\varepsilon_{iklm} A^k \dfrac{\partial^2 A^m}{\partial x_i \partial x_l} = \varepsilon_{ilkm} A^k \dfrac{\partial^2 A^m}{\partial x_l \partial x_i} =$

$$= \varepsilon_{i'k l' m} A^k \frac{\partial^2 A^m}{\partial x_{l'} \partial x_{i'}} = \begin{pmatrix} l' \to i \\ i' \to l \end{pmatrix} = \varepsilon_{lkim} A^k \frac{\partial^2 A^m}{\partial x_i \partial x_l}$$

However $\varepsilon_{iklm} = -\varepsilon_{lkim} \Rightarrow \varepsilon_{iklm} A^k \dfrac{\partial^2 A^m}{\partial x_i \partial x_l} = 0.$

$$\Rightarrow \quad \varepsilon_{iklm} F^{ik} F^{lm} = 4 \frac{\partial}{\partial x_i}\left(\varepsilon_{iklm} A^k \frac{\partial A^m}{\partial x_l}\right)$$

is full derivative.

For this reasons it can not be used for constructing action for EM field

$$\int d\Omega\, \varepsilon_{iklm} F^{ik} F^{lm} = 4 \int d\Omega \frac{\partial}{\partial x_i}\left(\varepsilon_{iklm} A^k \frac{\partial A^m}{\partial x_l}\right)$$

$$= 4 \int_{\Sigma} d\Sigma_i\, \varepsilon_{iklm} A^k \frac{\partial A^m}{\partial x_l}$$

complete surface term, no variation since all fields are fixed on the boundary Σ.

Define <u>Dual field tensor:</u> $\mathcal{F}_{\alpha\beta} = \varepsilon_{\alpha\beta\mu\nu} F^{\mu\nu}$

then $\varepsilon_{\alpha\beta\mu\nu} F^{\alpha\beta} F^{\mu\nu} = \mathcal{F}_{\alpha\beta} F^{\alpha\beta}$.

$$\mathcal{F}_{\alpha\beta} = \varepsilon_{\alpha\beta 01} F^{01} + \underbrace{\varepsilon_{\alpha\beta 10} F^{10}}_{-\varepsilon_{\alpha\beta 01}(-F^{01})} + 2\varepsilon_{\alpha\beta 02} F^{02} +$$

$$+2\varepsilon_{\alpha\beta 03} F^{03} + 2\varepsilon_{\alpha\beta 12} F^{12} + 2\varepsilon_{\alpha\beta 13} F^{13} +$$

$$+2\varepsilon_{\alpha\beta 23} F^{23} =$$

$$= 2\left\{\varepsilon_{\alpha\beta 01}\left(-\frac{E_x}{c}\right) + \varepsilon_{\alpha\beta 02}\left(-\frac{E_y}{c}\right) + \varepsilon_{\alpha\beta 03}\left(-\frac{E_z}{c}\right) + \varepsilon_{\alpha\beta 12}(-B_z) + \varepsilon_{\alpha\beta 13}(B_y) + \varepsilon_{\alpha\beta 23}(-B_x)\right\}$$

$$\Rightarrow \mathcal{F}_{01} = 2\underbrace{\varepsilon_{0123}}_{-1}(-B_x) = 2B_x$$

since all other elements have indices 0 or 1 appearing twice! $\Rightarrow$ give $\emptyset$.

$$\mathcal{F}_{02} = 2\underbrace{\varepsilon_{0213}}_{-\varepsilon_{0123}=+1} B_y = 2B_y$$

$$\mathcal{F}_{03} = 2\underbrace{\varepsilon_{0312}}_{\varepsilon_{0312} = -\varepsilon_{0132} = -(-\varepsilon_{0123}) = -1}(-B_z) = 2B_z$$

$$\mathcal{F}_{12} = 2\,\underbrace{\varepsilon_{1203}}_{-\varepsilon_{1023} = -(-\varepsilon_{0123}) = -1}\left(-\frac{E_z}{c}\right) = 2\,\frac{E_z}{c}$$

$$\mathcal{F}_{13} = 2\,\underbrace{\varepsilon_{1302}}_{-\varepsilon_{1032} = -(-\varepsilon_{0132}) = -\varepsilon_{0123} = +1}\left(-\frac{E_y}{c}\right) = -2\,\frac{E_y}{c}$$

$$\mathcal{F}_{23} = 2\,\underbrace{\varepsilon_{2301}}_{-\varepsilon_{2031} = -(-\varepsilon_{0231}) = -\varepsilon_{0213} = -(-\varepsilon_{0123}) = -1}\left(-\frac{E_x}{c}\right) = 2\,\frac{E_x}{c}$$

Hence we obtain

dual field tensor

$$\mathcal{F}_{\alpha\beta} = 2\begin{pmatrix} 0 & B_x & B_y & B_z \\ -B_x & 0 & E_z/c & -E_y/c \\ -B_y & -E_z/c & 0 & E_x/c \\ -B_z & E_y/c & -E_x/c & 0 \end{pmatrix} \Rightarrow \mathcal{F}^{\alpha\beta} = 2\begin{pmatrix} 0 & -B_x & -B_y & -B_z \\ B_x & 0 & E_z/c & -E_y/c \\ B_y & -E_z/c & 0 & E_x/c \\ B_z & E_y/c & -E_x/c & 0 \end{pmatrix}$$

while field tensor

$$F^{\alpha\beta} = \begin{pmatrix} 0 & -E_x/c & -E_y/c & -E_z/c \\ E_x/c & 0 & -B_z & B_y \\ E_y/c & B_z & 0 & -B_x \\ E_z/c & -B_y & B_x & 0 \end{pmatrix}$$

$\mathcal{F}^{\alpha\beta}$ obtains from $F^{\alpha\beta}$ by $\begin{cases} \frac{\vec{E}}{c} \to \vec{B}, \\ \vec{B} \to -\vec{E}. \end{cases}$

And finally we can find the invariant

$$\mathcal{F}_{\alpha\beta} F^{\alpha\beta} = \varepsilon_{\alpha\beta\mu\nu} F^{\alpha\beta} F^{\mu\nu} =$$

$$= \mathcal{F}_{01} F^{01} + \underbrace{\mathcal{F}_{10} F^{10}}_{(-\mathcal{F}_{01})(-F^{01})} + 2\mathcal{F}_{02} F^{02} + 2\mathcal{F}_{03} F^{03}$$

$$+ 2\mathcal{F}_{12} F^{12} + 2\mathcal{F}_{13} F^{13} + 2\mathcal{F}_{23} F^{23} =$$

$$= 4\left\{ B_x\left(-\frac{E_x}{c}\right) + B_y\left(-\frac{E_y}{c}\right) + B_z\left(-\frac{E_z}{c}\right) \right.$$

$$\left. + \frac{E_z}{c}(-B_z) + \left(-\frac{E_y}{c}\right) B_y + \frac{E_x}{c}(-B_x) \right\}$$

$$= -\frac{4}{c}\, \vec{E} \cdot \vec{B}$$

Variation of EM action

$$S_{int} + S_{field} = -\frac{1}{c}\int d\Omega\, A_\alpha j^\alpha - \frac{1}{4\mu_0 c}\int d\Omega\, F_{\alpha\beta}F^{\alpha\beta}$$

Now $A^\alpha(t,\vec{r}) \to A^\alpha(t,\vec{r}) + \delta A^\alpha(t,\vec{r})$

but $j^\alpha(t,\vec{r})$ is held fixed!

$$\Rightarrow \delta S_{int} = -\frac{1}{c}\int d\Omega\, j^\alpha \delta A_\alpha$$

$$\delta\left(F_{\alpha\beta}F^{\alpha\beta}\right) = \delta F_{\alpha\beta}\, F^{\alpha\beta} + F_{\alpha\beta}\,\delta F^{\alpha\beta} = 2F^{\alpha\beta}\delta F_{\alpha\beta} =$$

$$= 2F^{\alpha\beta}\left(\frac{\partial \delta A_\beta}{\partial x^\alpha} - \frac{\partial \delta A_\alpha}{\partial x^\beta}\right) = 2F^{\alpha\beta}\left(\frac{\partial \delta A_\beta}{\partial x^\alpha} + \frac{\partial \delta A_\alpha}{\partial x^\beta} - \right.$$

$$\left. - 2\frac{\partial \delta A_\alpha}{\partial x^\beta}\right) = 2\underset{\substack{\uparrow \\ \text{antisymmetric}}}{F^{\alpha\beta}}\underbrace{\left(\frac{\partial \delta A_\beta}{\partial x^\alpha} + \frac{\partial \delta A_\alpha}{\partial x^\beta}\right)}_{\text{symmetric}} - 4F^{\alpha\beta}\frac{\partial \delta A_\alpha}{\partial x^\beta}$$

$$\Downarrow$$

$$\phi$$

$$\Rightarrow \delta S_{field} = -\frac{1}{4\mu_0 c}(-4)\int d\Omega\, F^{\alpha\beta}\frac{\partial \delta A_\alpha}{\partial x^\beta} =$$

$$= \frac{1}{\mu_0 c}\int d\Omega\, \frac{\partial}{\partial x^\beta}\left(F^{\alpha\beta}\delta A_\alpha\right) - \frac{1}{\mu_0 c}\int d\Omega\, \frac{\partial F^{\alpha\beta}}{\partial x^\beta}\delta A_\alpha$$

$$= \frac{-1}{\mu_0 c}\int d\Omega\, \frac{\partial F^{\alpha\beta}}{\partial x^\beta}\delta A_\alpha + \underbrace{\frac{1}{\mu_0 c}\int_\Sigma dS_\beta\, F^{\alpha\beta}\delta A_\alpha}_{\phi}$$

because $\delta A_\alpha = 0$ on Σ

Thus $\delta S_{mat} + \delta S_{field} = -\frac{1}{c}\int d\Omega\, j^\alpha \delta A_\alpha - \frac{1}{\mu_0 c}\int d\Omega$

$$\cdot \frac{\partial F^{\alpha\beta}}{\partial x^\beta}\delta A_\alpha$$

$$\Rightarrow \quad j^\alpha + \frac{1}{\mu_0}\frac{\partial F^{\alpha\beta}}{\partial x^\beta} = 0$$

$$\boxed{\frac{\partial F^{\alpha\beta}}{\partial x^\beta} = -\mu_0 j^\alpha}$$ Lorentz invariant form!

$\alpha = 0$:
$$\frac{\partial F^{01}}{\partial x^1} + \frac{\partial F^{02}}{\partial x^2} + \frac{\partial F^{03}}{\partial x^3} = -\mu_0 j^0 = -\mu_0 c\rho$$

$$F^{0i} = -\frac{1}{c}E_i$$

$$\Rightarrow -\frac{1}{c}\frac{\partial E_i}{\partial x_i} = -\mu_0 c\rho$$

$$\vec{\nabla}\cdot\vec{E} = \mu_0 c^2 \rho = \frac{\rho}{\varepsilon_0}$$ Gauss law

$$\varepsilon_0 \mu_0 = c^2$$

dielectric permittivity of vacuum.

$\alpha = 1$:
$$\frac{\partial F^{10}}{\partial x^0} + \frac{\partial F^{12}}{\partial x^2} + \frac{\partial F^{13}}{\partial x^3} = -\mu_0 j^1$$

$$\frac{1}{c}\frac{\partial}{\partial t}\left(\frac{E_x}{c}\right) - \frac{\partial B_z}{\partial y} + \frac{\partial B_y}{\partial z} = -\mu_0 j_x$$

$$\left(\vec{\nabla}\times\vec{B}\right)_x = \left(\mu_0 \vec{j} + \frac{1}{c^2}\frac{\partial \vec{E}}{\partial t}\right)_x$$

$\Rightarrow \alpha = 1,2,3$ give us Ampere's law

$$\vec{\nabla} \times \vec{B} = \mu_0 \vec{j} + \frac{1}{c^2} \frac{\partial \vec{E}}{\partial t} .$$

displacement current (Maxwell)

Observe $\underbrace{\vec{\nabla} \cdot (\vec{\nabla} \times \vec{B})}_{\phi} = \mu_0 \vec{\nabla} \cdot \vec{j} + \frac{1}{c^2} \vec{\nabla} \cdot \frac{\partial \vec{E}}{\partial t}$

$$\Rightarrow \vec{\nabla} \cdot \vec{j} = -\frac{1}{\mu_0 c^2} \frac{\partial}{\partial t} (\vec{\nabla} \cdot \vec{E}) = -\varepsilon_0 \frac{\partial}{\partial t} \left(\frac{\rho}{\varepsilon_0} \right)$$

$$= -\frac{\partial \rho}{\partial t} .$$

$\Rightarrow$ displacement current is required by the charge conservation / continuity eqn.

Relativistically: $\frac{\partial j^\alpha}{\partial x^\alpha} = -\frac{1}{\mu_0} \frac{\partial^2 F^{\alpha\beta}}{\partial x^\alpha \partial x^\beta} = 0$ since $F^{\alpha\beta}$ is antisymmetric.

Where are the other two Maxwell's eqns?

$$\vec{\nabla}\cdot\vec{B} = 0 \quad \text{and} \quad \vec{\nabla}\times\vec{E} = -\frac{\partial \vec{B}}{\partial t}$$

Consider $F^{\alpha\beta} = \cancel{\frac{\partial A^\beta}{\partial x_\alpha} - \frac{\partial A^\alpha}{\partial x_\beta}}$ $\dfrac{\partial A^\beta}{\partial x_\alpha} - \dfrac{\partial A^\alpha}{\partial x_\beta}$

$$\frac{\partial F^{\alpha\beta}}{\partial x_\mu} = \frac{\partial^2 A^\beta}{\partial x_\mu \partial x_\alpha} - \frac{\partial A^\alpha}{\partial x_\mu \partial x_\beta}$$

Then $\dfrac{\partial F^{\alpha\beta}}{\partial x_\mu} + \dfrac{\partial F^{\mu\alpha}}{\partial x_\beta} + \dfrac{\partial F^{\beta\mu}}{\partial x_\alpha} = 0$

$\alpha \to \mu \to \beta \to \alpha$

because $\underline{\underline{\dfrac{\partial^2 A^\beta}{\partial x_\mu \partial x_\alpha}}} - \underline{\underline{\dfrac{\partial^2 A^\alpha}{\partial x_\mu \partial x_\beta}}} + \underline{\underline{\dfrac{\partial^2 A^\alpha}{\partial x_\beta \partial x_\mu}}} - \underline{\dfrac{\partial^2 A^\mu}{\partial x_\beta \partial x_\alpha}} +$

$$+ \underline{\frac{\partial^2 A^\mu}{\partial x_\alpha \partial x_\beta}} - \underline{\underline{\frac{\partial^2 A^\beta}{\partial x_\alpha \partial x_\mu}}} = 0$$

Note now that all indices must be different:

for if $\alpha = \beta \Rightarrow \dfrac{\partial F^{\alpha\alpha}}{\partial x_\mu} = 0$

$$\Rightarrow \frac{\partial F^{\mu\alpha}}{\partial x_\alpha} + \frac{\partial F^{\alpha\mu}}{\partial x_\alpha} = \frac{\partial}{\partial x_\alpha}\left(F^{\mu\alpha} + F^{\alpha\mu}\right) = 0$$

$\Rightarrow \alpha \neq \beta \neq \mu$

Let $\alpha = 0, \beta = 1, \mu = 2$:

$$\frac{\partial F^{01}}{\partial x_2} + \frac{\partial F^{20}}{\partial x_1} + \frac{\partial F^{12}}{\partial x_0} = 0$$

$$-\frac{\partial}{\partial y}\left(-\frac{E_x}{c}\right) - \frac{\partial}{\partial x}\left(E_y/c\right) + \frac{1}{c}\frac{\partial}{\partial t}\left(-B_z\right) = 0$$

$$\frac{\partial E_x}{\partial y} - \frac{\partial E_y}{\partial x} = \frac{\partial}{\partial t} B_z \Rightarrow \left(\vec{\nabla} \times \vec{E}\right)_z = -\left(\frac{\partial}{\partial t}\vec{B}\right)_z$$

$$\begin{vmatrix} \hat{x} & \hat{y} & \hat{z} \\ \partial_x & \partial_y & \partial_z \\ E_x & E_y & E_z \end{vmatrix} \Rightarrow \hat{z}\left(\frac{\partial E_y}{\partial x} - \frac{\partial E_x}{\partial y}\right)$$

Let $\alpha = 1, \beta = 2, \mu = 3$:

$$\frac{\partial F^{12}}{\partial x_3} + \frac{\partial F^{31}}{\partial x_2} + \frac{\partial F^{23}}{\partial x_1} = 0$$

$$-\frac{\partial}{\partial z}\left(-B_z\right) - \frac{\partial}{\partial y}\left(-B_y\right) - \frac{\partial}{\partial x}\left(-B_x\right) = 0$$

$$\vec{\nabla} \cdot \vec{B} = 0.$$

$\rightarrow$ All Maxwell's equations are obtained from $S_{field} + S_{int}$!